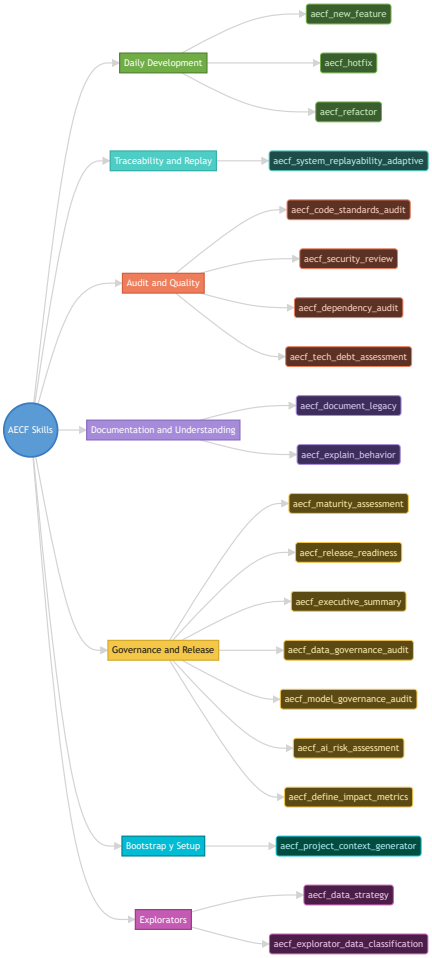
AECF Skills — Catalog by Use Category

Quick guide to know **which skill to use** depending on the situation. Each skill is classified by its main purpose.

Visual Summary



Category	Skills
🔧 **Desarrollo Diario**	`aecf_new_feature` · `aecf_hotfix` · `aecf_refactor`
🔄 **Trazabilidad y Replay**	`aecf_system_replayability_adaptive`
🔍 **Audit and Quality**	`aecf_code_standards_audit` · `aecf_security_review` · `aecf_dependency_audit` · `aecf_tech_debt_assessment`
📚 **Documentation and Understanding**	`aecf_document_legacy` · `aecf_explain_behavior`
🏛️ **Governance and Release**	`aecf_maturity_assessment` · `aecf_release_readiness` · `aecf_executive_summary` · `aecf_data_governance_audit` · `aecf_model_governance_audit` · `aecf_ai_risk_assessment` · `aecf_define_impact_metrics`
🚀 **Bootstrap y Setup**	`aecf_project_context_generator`
🔬 **Explorators**	`aecf_explorator_data_classification` · `aecf_data_strategy`

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1. 🔧 Daily Development

Skills that are used in the day-to-day work of generating, modifying and implementing code.

Skill	ID	Description	When to Use It
New Feature	`aecf_new_feature`	Complete flow to implement new functionality with testing	New feature, endpoint, module or component from scratch
Hotfix	`aecf_hotfix`	Accelerated response to critical production incidents	Production down (P1), vulnerability exploited, data loss active, core functionality degraded (P2)

Refactor	`aecf_refactor`	Governed refactoring with prior documentation and regression checking	Post-standards audit, reduce cyclomatic complexity, eliminate duplication, prepare legacy code for new feature
System Replayability	`aecf_system_replayability_adaptive`	Replay/traceability adaptive to any architecture	Add reproducibility/replay to a project, traceability for compliance, post-document_legacy, post-security_review, pre-release_readiness

Typical development flow

```
aecf_new_feature (implementar)
  → aecf_release_readiness (validate before release)
```

Typical flow of improving existing code

```
aecf_document_legacy → aecf_code_standards_audit → aecf_refactor
```

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2. 🔍 Audit and Quality

read-only analysis skills that generate reports without modifying code. Ideal for periodic evaluations, pre-release or due diligence.

Unified group characteristics:

- 🗂️ **Index of Sections Analyzed**: Before the findings, each report includes an index with direct links to each section analyzed and its remediation
- 🔧 **Remediation by Skills**: Each CRITICAL/WARNING finding recommends the **appropriate AECF skill** (never prompts/internal phases directly, to maintain the cycle of predictability)
- 🔄 **Matrix Auto-Apply Protocol**: `ADD_RULE` decisions are automatically applied to the project severity matrix — without manual intervention, with full traceability to the source report

Skill	ID	Description	When to Use It
Code Standards Audit	`aecf_code_standards_audit`	Audits compliance with standards defined in GLOBAL_CONTEXT and PROJECT_CONTEXT	Audit legacy code, verify organizational conventions, pre-refactoring, onboarding new code to the project
Security Review	`aecf_security_review`	Specialized security audit with CVSS classification	Pre-deployment, code that handles sensitive data, post-security compromise, compliance (PCI-DSS, HIPAA, GDPR)
Dependency Audit	`aecf_dependency_audit`	Dependency audit: CVEs, licenses, maintenance health, supply chain	Pre-release, periodic evaluation (monthly/quarterly), Dependabot/Snyk post-alert, project due diligence
Tech Debt Assessment	`aecf_tech_debt_assessment`	Comprehensive technical debt assessment with quantification and prioritization	Before major refactoring, sprint planning, technical due diligence, pre-migration, justify investment to stakeholders

Unified output format

All skills in this group generate reports with:

```
📊 Executive Summary
🗂️ Sections Analyzed — Navigation Index ← NEW
(direct links to each section with findings)
📋 Findings by Section
   - Cada CRITICAL/WARNING → 🔧 Skill recomendado: `skill: aecf_xxx`
- NEVER refers to internal prompts/phases
📈 Metrics / Scoring
🎯 Recommendations
```

Recommended audit chains

```
aecf_code_standards_audit → aecf_tech_debt_assessment → aecf_refactor
aecf_security_review → aecf_dependency_audit (drill down into third parties)
aecf_tech_debt_assessment → aecf_maturity_assessment (evaluar estado global)
```

Suggested frequency

Audit	Recommended Frequency
Code Standards Audit	Cada sprint o pre-refactoring
Security Review	Pre-release and quarterly
Dependency Audit	Monthly or upon security alerts

Tech Debt Assessment Quarterly or pre-planning

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3. 📚 Documentation and Understanding

Skills to understand and document existing systems. Essential for onboarding, change preparation and incident analysis.

Skill	ID	Description	When to Use It
Legacy Document	`aecf_document_legacy`	Document existing legacy code with flowcharts	Before modifying code without documentation, onboarding new equipment, understanding legacy system, preparing refactoring
Explain Behavior	`aecf_explain_behavior`	Analyze and explain why the system behaves in a certain way	Understand unexpected outputs, investigate 403 errors/timeouts/incorrect calculations, pre-debugging preparation, analyze critical flows

Key differences

Aspecto	`document_legacy`	`explain_behavior`
Focus	What the code does (general functionality)	Why it produces a certain output (specific behavior)
Output	Complete documentation + `.mmd` diagrams	Behavior traceability analysis
Next step	Refactor, new feature, audit	Debug, fix plan, or nothing if correct

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4. 🏛️ Governance and Release

Skills to evaluate process maturity and validate preparation for releases. Oriented towards technical leadership and compliance.

Skill	ID	Description	When to Use It
Maturity Assessment	`aecf_maturity_assessment`	Evaluates 10 dimensions of AECF governance, calculates level L1–L5, generates roadmap	Initial adoption of AECF (baseline), quarterly review, post-incident, pre-certification/audit, team onboarding
Release Readiness	`aecf_release_readiness`	Comprehensive pre-release validation with GO/NO-GO verdict	Before each release to production, pre-merge to main, sprint end, compliance checkpoint, before tagging version
Executive Summary	`aecf_executive_summary`	Generates consolidated executive summary of all documents of a TOPIC	When an overall executive summary is required from `documentation/{{TOPIC}}/`; Mandatory explicit TOPIC
Data Governance Audit	`aecf_data_governance_audit`	Data classification, lineage, retention and controls audit	Validate data traceability, retention compliance, pre-release and audit preparation
Model Governance Audit	`aecf_model_governance_audit`	Model impact audit, inference paths and auditability	Changes that alter behavior AI, validation of determinism and change control
AI Risk Assessment	`aecf_ai_risk_assessment`	AI risk assessment (operational, security, compliance, reliability)	Pre-go-live, post-incident, executive exposure assessment and mitigations
Define Impact Metrics	`aecf_define_impact_metrics`	Define metrics with baseline, targets and measurement governance	When you need to measure real impact of change with auditable criteria

When to use each

Situation	Skill
"Are we ready to deploy?"	`aecf_release_readiness`
"How mature is our process?"	`aecf_maturity_assessment`
"What are we missing as a team?"	`aecf_maturity_assessment`
"Did all the audits for this feature pass?"	`aecf_release_readiness`
"Generate an executive summary of the TOPIC"	`aecf_executive_summary`
“Do we have data lineage and retention well controlled?”	`aecf_data_governance_audit`
"Does this change affect inference or model behavior?"	`aecf_model_governance_audit`
"What is our current AI risk and how to mitigate it?"	`aecf_ai_risk_assessment`
"How do we measure impact with baseline and goals?"	`aecf_define_impact_metrics`
"What PII/SENSITIVE fields exist in the project?"	`aecf_explorator_data_classification`
"I need a classified data inventory before auditing"	`aecf_explorator_data_classification`

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5. 🔍 Explorators

Knowledge Extraction skills that generate structured documentation to feed audit, enforcement and fix skills. Explorators are always READ-ONLY on the source code.

Features:

- 🔍 **Structured discovery**: Inspect project artifacts without modifying them
- 📄 **Deterministic output**: Generate YAML structured and validated by schema
- 🔒 **Read-only**: NEVER modify code or data models
- 🔗 **Enablers**: Their output is consumed directly by audit, enforcement and fix skills

Skill	ID	Description	When to Use It
Explorer Data Classification	`aecf_explorator_data_classification`	Discover all the data models, schemas, migrations and DTOs of the project. Apply deterministic classification heuristics (PII/SENSITIVE/CONFIDENTIAL/INTERNAL/PUBLIC) and generate structured governance proposal in YAML	Pre-data audit, before `aecf_data_governance_audit`, project onboarding with data, GDPR/CCPA compliance, pre-release with PII
Data Strategy	`aecf_data_strategy`	Design of data ingestion, storage and management strategy	Integrate high volume data source, decide ETL/ELT strategy, design deduplication, pre-input for features with massive data

Recommended usage flow (Explorators)

```
aecf_explorator_data_classification (discover and classify)
  → aecf_data_strategy (define ingestion and storage strategy)
  → aecf_data_governance_audit (auditar gaps usando el inventario)
→ aecf_ai_risk_assessment (assess risk on classified data)
→ aecf_release_readiness (validate before release)
```

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6. 🚀 Bootstrap y Setup

Skills to initialize and configure the AECF environment in a project. They must be executed **before** any other skill.

Skill	ID	Description	When to Use It
Project Context Generator	`aecf_project_context_generator`	Scans the entire workspace and generates `_00_PROJECT_CONTEXT.md` automatically	When starting to use AECF in a project, when PROJECT_CONTEXT does not exist, legacy project onboarding, update outdated context

Recommended AECF adoption flow

```
aecf_project_context_generator (PRIMERO — contexto base)
  → aecf_maturity_assessment (evaluar estado)
  → aecf_code_standards_audit (auditar)
→ Development skills as needed
```

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6. 🧭 Quick Decision Guide

"What skill do I need?"

```
What do you need to do?
│
├─ Get started with AECF in a project
│  └─ Generate project context → aecf_project_context_generator
│
├─ Write new code
│  ├─ Feature nueva → aecf_new_feature
│  ├─ Urgent production fix → aecf_hotfix
│  ├─ Improve existing code → aecf_refactor
│  └─ Add replay/traceability → aecf_system_replayability_adaptive
│
├─ Evaluate/audit without touching code
│  ├─ Quality and standards → aecf_code_standards_audit
│  ├─ Security → aecf_security_review
│  ├─ Dependencies/third parties → aecf_dependency_audit
│  └─ Technical Debt → aecf_tech_debt_assessment
│
├─ Understand/document
│  ├─ Document old code → aecf_document_legacy
│  └─ Explain behavior → aecf_explain_behavior
│
```

```
├─ Govern/validate
│  ├─ Ready for release? → aecf_release_readiness
│  ├─ How mature are we? → aecf_maturity_assessment
│  ├─ Do we comply with data governance? → aecf_data_governance_audit
│  ├─ Is there model impact? → aecf_model_governance_audit
│  ├─ What is the AI risk? → aecf_ai_risk_assessment
│  └─ How do we measure impact? → aecf_define_impact_metrics
│
└─ Explore/discover data
└─ What PII/SENSITIVE data does the project have? → aecf_explorator_data_classification
└─ Need ingestion/storage strategy for high-volume data? → aecf_data_strategy
```

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7. 📋 Complete Skills Matrix

Skill	Modify Code	Generate Docs	Estimated Time	Mode
`aecf_project_context_generator`	❌	✅	5 – 15min	Bootstrap
`aecf_new_feature`	✅	✅	1.5 – 4h	Execution
`aecf_hotfix`	✅	✅	1 – 3h	Execution
`aecf_refactor`	✅	✅	2 – 6h	Execution
`aecf_data_strategy`	❌	✅	20min – 3h	Explorator
`aecf_system_replayability_adaptive`	✅	✅	1.5 – 6h	Execution
`aecf_code_standards_audit`	❌	✅	30min – 4h	Read-only
`aecf_security_review`	❌	✅	45min – 6h	Read-only
`aecf_dependency_audit`	❌	✅	30min – 2h	Read-only
`aecf_tech_debt_assessment`	❌	✅	1 – 4h	Read-only
`aecf_document_legacy`	❌	✅	30min – 2h	Read-only
`aecf_explain_behavior`	❌	✅	15 – 60min	Read-only
`aecf_maturity_assessment`	❌	✅	1 – 3h	Read-only
`aecf_release_readiness`	❌	✅	30min – 2h	Read-only
`aecf_data_governance_audit`	❌	✅	30min – 3h	Read-only
`aecf_model_governance_audit`	❌	✅	30min – 3h	Read-only
`aecf_ai_risk_assessment`	❌	✅	30min – 3h	Read-only
`aecf_define_impact_metrics`	❌	✅	20min – 2h	Read-only
`aecf_explorator_data_classification`	❌	✅	15min – 2h	Explorator

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8. 🔗 Frequent Compositions

Bootstrap + full AECF adoption

```
aecf_project_context_generator → aecf_maturity_assessment → aecf_code_standards_audit
```

Complete new feature cycle

```
aecf_new_feature → aecf_security_review → aecf_release_readiness
```

Legacy code rescue

```
aecf_document_legacy → aecf_code_standards_audit → aecf_tech_debt_assessment → aecf_refactor
```

Comprehensive pre-release audit

```
aecf_code_standards_audit → aecf_security_review → aecf_dependency_audit → aecf_release_readiness
```

Incident response

```
aecf_hotfix → aecf_security_review → aecf_explain_behavior (post-mortem)
```

Legacy project evaluation

```
aecf_document_legacy → aecf_tech_debt_assessment → aecf_dependency_audit → aecf_maturity_assessment
```

Big Data Pipeline

```
aecf_data_strategy → aecf_new_feature → aecf_security_review → aecf_release_readiness
```

Add traceability/replay to existing project

```
aecf_document_legacy → aecf_system_replayability_adaptive → aecf_security_review → aecf_release_readiness
```

Comprehensive governance block

```
aecf_data_governance_audit → aecf_model_governance_audit → aecf_ai_risk_assessment → aecf_define_impact_metrics
```

Complete data exploration and auditing

```
aecf_explorator_data_classification → aecf_data_governance_audit → aecf_ai_risk_assessment → aecf_release_readiness
```

> **Tip**: All skills are invoked with `skill: <name>` or natural language. The SKILL_DISPATCHER auto-resolves parameters, TOPIC and numbering.

GOVERNANCE VALIDATION BLOCK

- Data lineage impact
- Model impact (YES/NO)
- Risk impact
- Compliance check